



International Data Encryption Algorithm (IDEA)

LESSON 28 OF 40



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CRYPTOGRAPHY & NETWORK SECURITY

International Data Encryption Algorithm (IDEA)

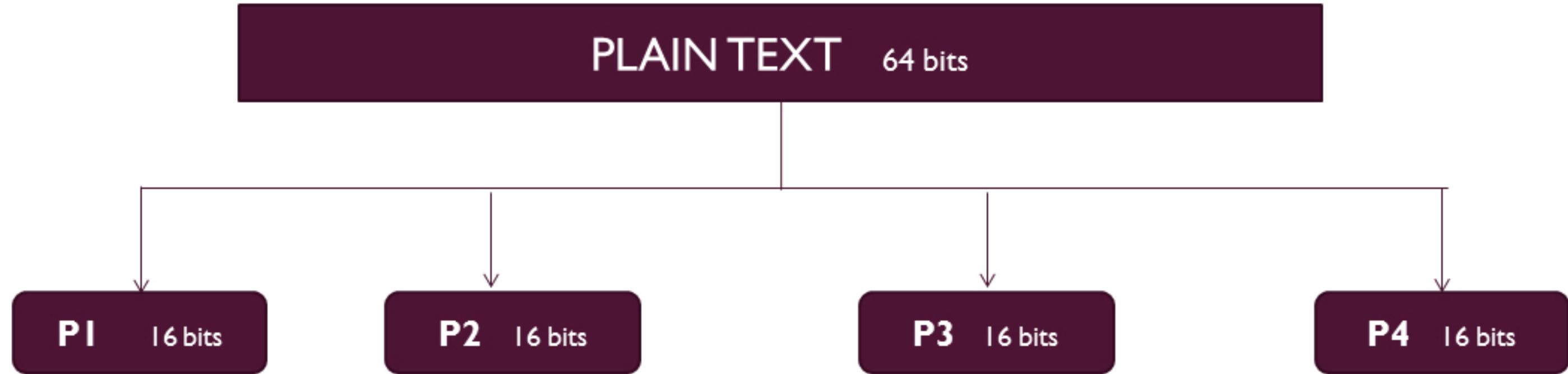
International Data Encryption Algorithm (IDEA)

- Symmetric block cipher.
- 64 bit block data is used.
- 128 bit key and 52 subkeys are used.
- 8 rounds.
- 6 keys used in each round except the last round.
- In last round 6 + 4 keys are used.
- 3 operations are used ;Addition, Multiplication and XOR.

SUBKEY GENERATION

- 128 bit key is divided into 8 parts of 16 bits each.
- Then a cyclic left shift is made for 25 positions to get a new 128 bit key.
- The process is repeated until required keys are generated.

ENCRYPTION



ONE ROUND OPERATIONS

- | | |
|----------------------------|----------------------------|
| 1. Multiply P1 and K1 | i.e, $S1 = P1 \times K1$ |
| 2. Add P2 and K2 | i.e, $S2 = P2 + K2$ |
| 3. Add P3 and K3 | i.e, $S3 = P3 + K3$ |
| 4. Multiply P4 and K4 | i.e, $S4 = P4 \times K4$ |
| 5. XOR step 1 and step 3 | i.e, $S5 = S1 \oplus S3$ |
| 6. XOR step 2 and step 4 | i.e, $S6 = S2 \oplus S4$ |
| 7. Multiply step 5 with K5 | i.e, $S7 = S5 \times K5$ |
| 8. Add step 6 and step 7 | i.e, $S8 = S6 + S7$ |
| 9. Multiply step 8 with K6 | i.e, $S9 = S8 \times K6$ |
| 10. Add step 7 and step 9 | i.e, $S10 = S7 + S9$ |
| 11. XOR step 1 and step 9 | i.e, $S11 = S1 \oplus S9$ |
| 12. XOR step 3 and step 9 | i.e, $S12 = S3 \oplus S9$ |
| 13. XOR step 2 and step 10 | i.e, $S13 = S2 \oplus S10$ |
| 14. XOR step 4 and step 10 | i.e, $S14 = S4 \oplus S10$ |

ONE ROUND OPERATIONS

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| 1. Multiply P1 and K1 | i.e, $S1 = P1 \times K1$ |
| 2. Add P2 and K2 | i.e, $S2 = P2 + K2$ |
| 3. Add P3 and K3 | i.e, $S3 = P3 + K3$ |
| 4. Multiply P4 and K4 | i.e, $S4 = P4 \times K4$ |
| 5. XOR step 1 and step 3 | i.e, $S5 = S1 \oplus S3$ |
| 6. XOR step 2 and step 4 | i.e, $S6 = S2 \oplus S4$ |
| 7. Multiply step 5 with K5 | i.e, $S7 = S5 \times K5$ |
| 8. Add step 6 and step 7 | i.e, $S8 = S6 + S7$ |
| 9. Multiply step 8 with K6 | i.e, $S9 = S8 \times K6$ |
| 10. Add step 7 and step 9 | i.e, $S10 = S7 + S9$ |
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ONE ROUND OPERATIONS

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| 2. Add P2 and K2 | i.e, $S2 = P2 + K2$ | |
| 3. Add P3 and K3 | i.e, $S3 = P3 + K3$ | |
| 4. Multiply P4 and K4 | i.e, $S4 = P4 \times K4$ | |
| 5. XOR step 1 and step 3 | i.e, $S5 = S1 \oplus S3$ | |
| 6. XOR step 2 and step 4 | i.e, $S6 = S2 \oplus S4$ | |
| 7. Multiply step 5 with K5 | i.e, $S7 = S5 \times K5$ | |
| 8. Add step 6 and step 7 | i.e, $S8 = S6 + S7$ | |
| 9. Multiply step 8 with K6 | i.e, $S9 = S8 \times K6$ | |
| 10. Add step 7 and step 9 | i.e, $S10 = S7 + S9$ | |
| 11. XOR step 1 and step 9 | i.e, $S11 = S1 \oplus S9$ | —————→ New P1 |
| 12. XOR step 3 and step 9 | i.e, $S12 = S3 \oplus S9$ | —————→ New P2 |
| 13. XOR step 2 and step 10 | i.e, $S13 = S2 \oplus S10$ | —————→ New P3 |
| 14. XOR step 4 and step 10 | i.e, $S14 = S4 \oplus S10$ | —————→ New P4 |

ONE ROUND OPERATIONS

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|----------------------------|----------------------------|---------------|
| 1. Multiply P1 and K1 | i.e, $S1 = P1 \times K1$ | |
| 2. Add P2 and K2 | i.e, $S2 = P2 + K2$ | |
| 3. Add P3 and K3 | i.e, $S3 = P3 + K3$ | |
| 4. Multiply P4 and K4 | i.e, $S4 = P4 \times K4$ | |
| 5. XOR step 1 and step 3 | i.e, $S5 = S1 \oplus S3$ | |
| 6. XOR step 2 and step 4 | i.e, $S6 = S2 \oplus S4$ | |
| 7. Multiply step 5 with K5 | i.e, $S7 = S5 \times K5$ | |
| 8. Add step 6 and step 7 | i.e, $S8 = S6 + S7$ | |
| 9. Multiply step 8 with K6 | i.e, $S9 = S8 \times K6$ | |
| 10. Add step 7 and step 9 | i.e, $S10 = S7 + S9$ | |
| 11. XOR step 1 and step 9 | i.e, $S11 = S1 \oplus S9$ | —————→ New P1 |
| 12. XOR step 3 and step 9 | i.e, $S12 = S3 \oplus S9$ | —————→ New P2 |
| 13. XOR step 2 and step 10 | i.e, $S13 = S2 \oplus S10$ | —————→ New P3 |
| 14. XOR step 4 and step 10 | i.e, $S14 = S4 \oplus S10$ | —————→ New P4 |
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I - 7 ROUND OPERATIONS

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|----------------------------|----------------------------|---------------|
| 1. Multiply P1 and K1 | i.e, $S1 = P1 \times K1$ | |
| 2. Add P2 and K2 | i.e, $S2 = P2 + K2$ | |
| 3. Add P3 and K3 | i.e, $S3 = P3 + K3$ | |
| 4. Multiply P4 and K4 | i.e, $S4 = P4 \times K4$ | |
| 5. XOR step 1 and step 3 | i.e, $S5 = S1 \oplus S3$ | |
| 6. XOR step 2 and step 4 | i.e, $S6 = S2 \oplus S4$ | |
| 7. Multiply step 5 with K5 | i.e, $S7 = S5 \times K5$ | |
| 8. Add step 6 and step 7 | i.e, $S8 = S6 + S7$ | |
| 9. Multiply step 8 with K6 | i.e, $S9 = S8 \times K6$ | |
| 10. Add step 7 and step 9 | i.e, $S10 = S7 + S9$ | |
| 11. XOR step 1 and step 9 | i.e, $S11 = S1 \oplus S9$ | —————→ New P1 |
| 12. XOR step 3 and step 9 | i.e, $S12 = S3 \oplus S9$ | —————→ New P3 |
| 13. XOR step 2 and step 10 | i.e, $S13 = S2 \oplus S10$ | —————→ New P2 |
| 14. XOR step 4 and step 10 | i.e, $S14 = S4 \oplus S10$ | —————→ New P4 |

8th ROUND OPERATIONS

- | | |
|-----------------------------|----------------------------|
| 1. Multiply P1 and K43 | i.e, $S1 = P1 \times K43$ |
| 2. Add P2 and K44 | i.e, $S2 = P2 + K44$ |
| 3. Add P3 and K45 | i.e, $S3 = P3 + K45$ |
| 4. Multiply P4 and K46 | i.e, $S4 = P4 \times K46$ |
| 5. XOR step 1 and step 3 | i.e, $S5 = S1 \oplus S3$ |
| 6. XOR step 2 and step 4 | i.e, $S6 = S2 \oplus S4$ |
| 7. Multiply step 5 with K47 | i.e, $S7 = S5 \times K47$ |
| 8. Add step 6 and step 7 | i.e, $S8 = S6 + S7$ |
| 9. Multiply step 8 with K48 | i.e, $S9 = S8 \times K48$ |
| 10. Add step 7 and step 9 | i.e, $S10 = S7 + S9$ |
| 11. XOR step 1 and step 9 | i.e, $S11 = S1 \oplus S9$ |
| 12. XOR step 3 and step 9 | i.e, $S12 = S3 \oplus S9$ |
| 13. XOR step 2 and step 10 | i.e, $S13 = S2 \oplus S10$ |
| 14. XOR step 4 and step 10 | i.e, $S14 = S4 \oplus S10$ |

8th ROUND OPERATIONS

- | | | |
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| 1. Multiply P1 and K43 | i.e, $S1 = P1 \times K43$ | |
| 2. Add P2 and K44 | i.e, $S2 = P2 + K44$ | |
| 3. Add P3 and K45 | i.e, $S3 = P3 + K45$ | |
| 4. Multiply P4 and K46 | i.e, $S4 = P4 \times K46$ | |
| 5. XOR step 1 and step 3 | i.e, $S5 = S1 \oplus S3$ | |
| 6. XOR step 2 and step 4 | i.e, $S6 = S2 \oplus S4$ | |
| 7. Multiply step 5 with K47 | i.e, $S7 = S5 \times K47$ | |
| 8. Add step 6 and step 7 | i.e, $S8 = S6 + S7$ | |
| 9. Multiply step 8 with K48 | i.e, $S9 = S8 \times K48$ | |
| 10. Add step 7 and step 9 | i.e, $S10 = S7 + S9$ | |
| 11. XOR step 1 and step 9 | i.e, $S11 = S1 \oplus S9$ | —————> New P1 |
| 12. XOR step 3 and step 9 | i.e, $S12 = S3 \oplus S9$ | —————> New P2 |
| 13. XOR step 2 and step 10 | i.e, $S13 = S2 \oplus S10$ | —————> New P3 |
| 14. XOR step 4 and step 10 | i.e, $S14 = S4 \oplus S10$ | —————> New P4 |

8th ROUND OPERATIONS

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| 9. Multiply step 8 with K48 | i.e, $S9 = S8 \times K48$ | |
| 10. Add step 7 and step 9 | i.e, $S10 = S7 + S9$ | |
| 11. XOR step 1 and step 9 | i.e, $S11 = S1 \oplus S9$ | → New P1 |
| 12. XOR step 3 and step 9 | i.e, $S12 = S3 \oplus S9$ | → New P2 |
| 13. XOR step 2 and step 10 | i.e, $S13 = S2 \oplus S10$ | → New P3 |
| 14. XOR step 4 and step 10 | i.e, $S14 = S4 \oplus S10$ | → New P4 |

No swapping



8th ROUND OPERATIONS

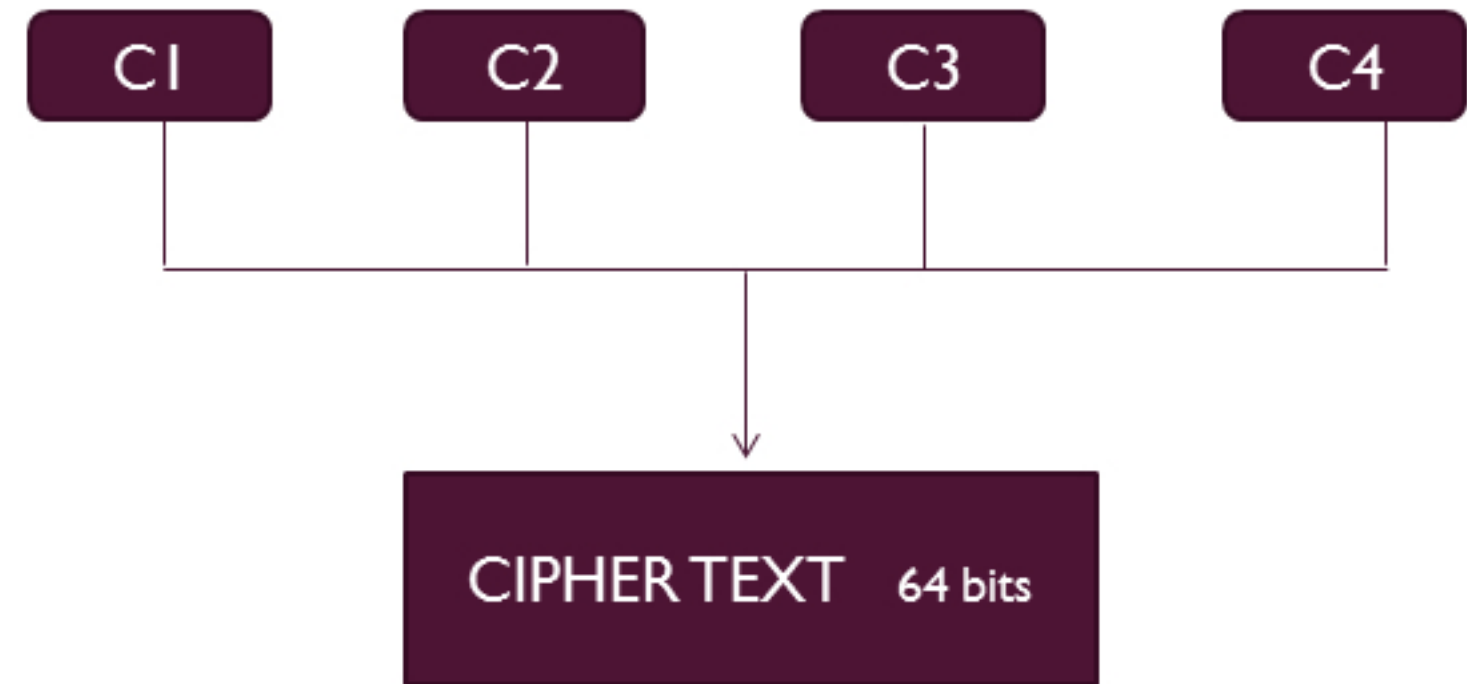
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| 13. XOR step 2 and step 10 | i.e, $S13 = S2 \oplus S10$ | → New P3 |
| 14. XOR step 4 and step 10 | i.e, $S14 = S4 \oplus S10$ | → New P4 |
| 15. Multiply P1 with K49 | i.e, $P1 \times K49$ | |
| 16. Add P2 with K50 | i.e, $P2 + K50$ | |
| 17. Add P3 with K51 | i.e, $P3 + K51$ | |
| 18. Multiply P4 with K52 | i.e, $P4 \times K52$ | |

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| 11. XOR step 1 and step 9 | i.e, $S11 = S1 \oplus S9$ |
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| 13. XOR step 2 and step 10 | i.e, $S13 = S2 \oplus S10$ |
| 14. XOR step 4 and step 10 | i.e, $S14 = S4 \oplus S10$ |
| 15. Multiply P1 with K49 | i.e, $P1 \times K49$ |
| 16. Add P2 with K50 | i.e, $P2 + K50$ |
| 17. Add P3 with K51 | i.e, $P3 + K51$ |
| 18. Multiply P4 with K52 | i.e, $P4 \times K52$ |

8th ROUND OPERATIONS

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| 13. XOR step 2 and step 10 | i.e, $S13 = S2 \oplus S10$ |
| 14. XOR step 4 and step 10 | i.e, $S14 = S4 \oplus S10$ |
| 15. Multiply P1 with K49 | i.e, $P1 \times K49 = C1$ |
| 16. Add P2 with K50 | i.e, $P2 + K50 = C2$ |
| 17. Add P3 with K51 | i.e, $P3 + K51 = C3$ |
| 18. Multiply P4 with K52 | i.e, $P4 \times K52 = C4$ |



DECRYPTION

- Decryption process is similar to encryption algorithm.
- But the keys will be used in reverse order and inverse of the keys are applied.

Next to Study :

Reference : Cryptography and Network Security Principles and Practices, Fourth Edition
By William Stallings

THANK YOU

PRESENTED BY,
ANSHA PK